Compressor off-design report: summary and diagnostics

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Rotor-Rotor Different Shaft  |  run: Max_PR_Compressor_design_rr_diff_shaft

Rotor-Rotor Different Shaft  |  PR_des = 2.4010, eta_des = 0.9239  |  correlations: polynomial, Howell factor 2.00

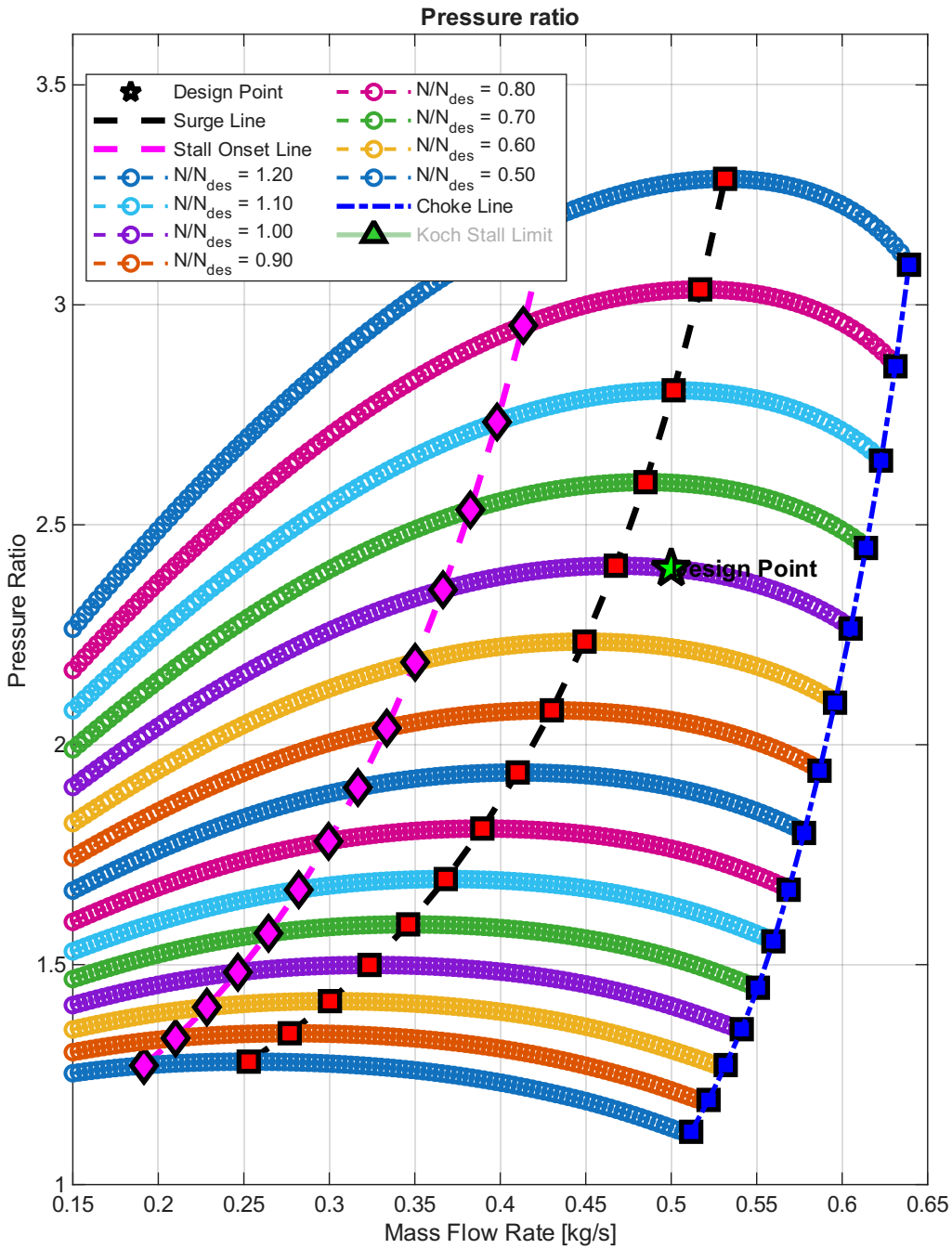
STABILITY (design speed): first limit = peak at m = 0.4676 kg/s (PR 2.407); stall margin SM = 7.2 %
   peak of the speed line (zero slope): m = 0.4676 kg/s | Howell stall onset: m = 0.3665 kg/s
   Koch stage limit: m = 0.5086 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.6050 kg/s (Rotor 1 exit (relative)) -> choke margin 21.0 %, operating range (m_choke - m_lim)/m_choke = 22.7 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.331 -> surge if the installation has V_p/(A_c L_c) > 5.85 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Rotor 1 (omega/omega_des = 1.06), lowest de Haller Rotor 1 (0.63). Details: Tables > Blade
      rows.

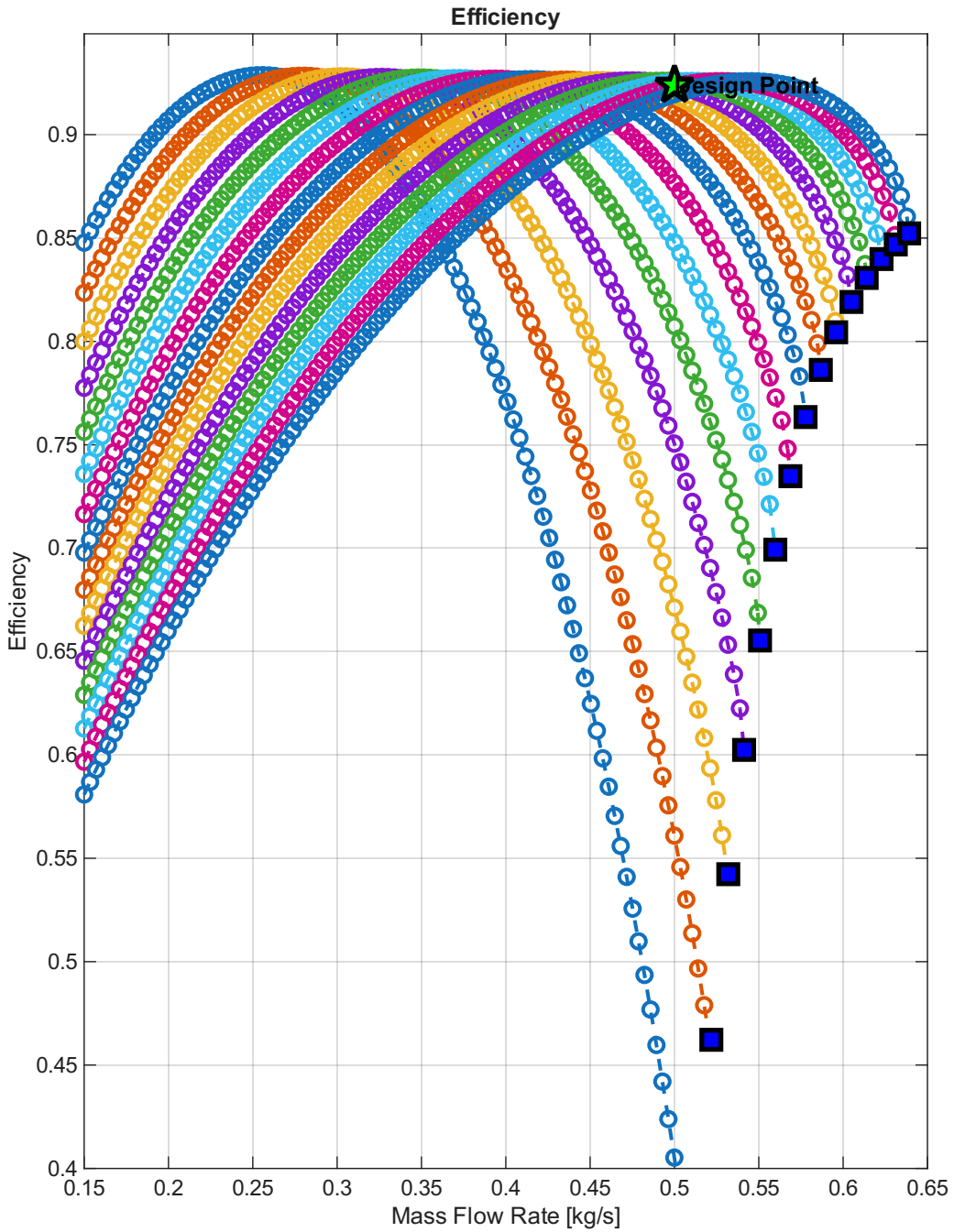
KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 15 of 15 speed lines (N/N_des 0.50 to 1.20).
   why: R2: effective dynamic-head factor F_ef = 0.35, outside 0.39-1.11. The upstream wakes enter in a frame turning the same way at a similar
      speed; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: BEYOND THE LIMIT AT DESIGN (1.03)
     - solidity alone cannot reach the target (L/g2 > 2.8 would be needed)
     - or LOWER the stage static-pressure rise by about 17% at the same velocities (lower loading psi or degree of reaction, or a larger flow
       coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.94
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.96
     - R2: effective dynamic-head factor F_ef = 0.35 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
       1 the ratio would be 0.74 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: F_ef = [0.96 0.35] (data 0.39-1.11) -> indicative only
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Inputs of this run (.mat data and settings)

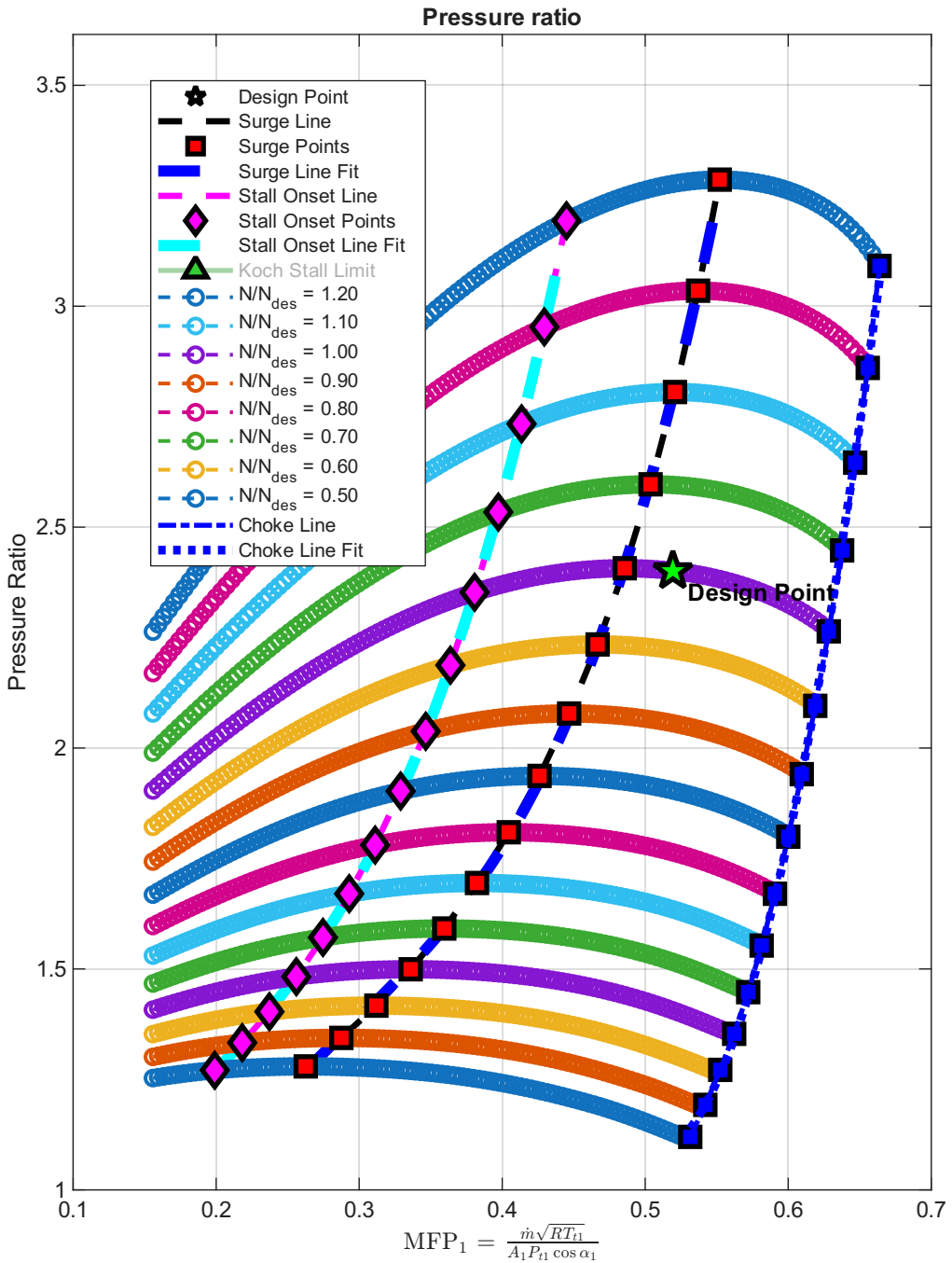
Variable	Value	Unit	Source
Configuration	Rotor-Rotor different shaft		
Input source	.mat: Max_PR_Compressor_design_rr_d...		
alpha1	22.2856	deg	.mat: Max_PR_Compressor_design_rr_d...
n_stages	2	-	.mat: Max_PR_Compressor_design_rr_d...
A_vec	0.00299142 0.00268389 0.00253739	m^2	.mat: Max_PR_Compressor_design_rr_d...
beta1_dash	45.3733	deg	.mat: Max_PR_Compressor_design_rr_d...
beta2_dash	17.2902	deg	.mat: Max_PR_Compressor_design_rr_d...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rr_d...
w_loss_rotor	0.118639	-	.mat: Max_PR_Compressor_design_rr_d...
beta3_dash	38.2869	deg	.mat: Max_PR_Compressor_design_rr_d...
beta4_dash	-17.8442	deg	.mat: Max_PR_Compressor_design_rr_d...
deviation_1	4.99536	deg	.mat: Max_PR_Compressor_design_rr_d...
deviation_2	8.53425	deg	.mat: Max_PR_Compressor_design_rr_d...
incidence_1	5.50411	deg	.mat: Max_PR_Compressor_design_rr_d...
incidence_2	3.75221	deg	.mat: Max_PR_Compressor_design_rr_d...
deflection_1	28.5918	deg	.mat: Max_PR_Compressor_design_rr_d...
deflection_2	51.349	deg	.mat: Max_PR_Compressor_design_rr_d...
w_loss_rotor2	0.110832	-	.mat: Max_PR_Compressor_design_rr_d...
U1	257.281	m/s	.mat: Max_PR_Compressor_design_rr_d...
U2	334.465	m/s	.mat: Max_PR_Compressor_design_rr_d...
sigma_1	1.96031	-	.mat: Max_PR_Compressor_design_rr_d...
sigma_2	1.95306	-	.mat: Max_PR_Compressor_design_rr_d...
chord	0.01	m	.mat: Max_PR_Compressor_design_rr_d...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	143	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

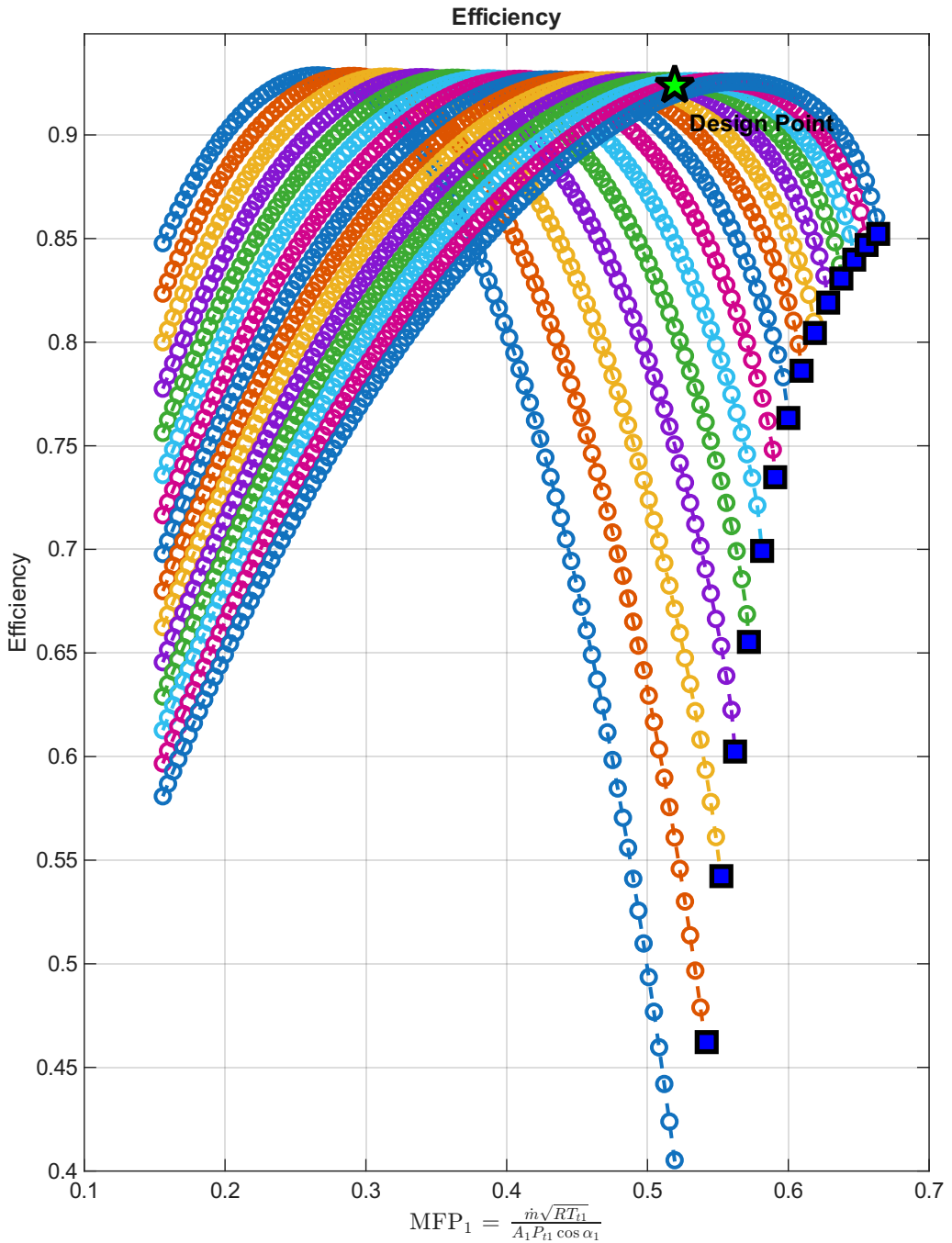
Standard (dimensional) map - Rotor-Rotor Different Shaft



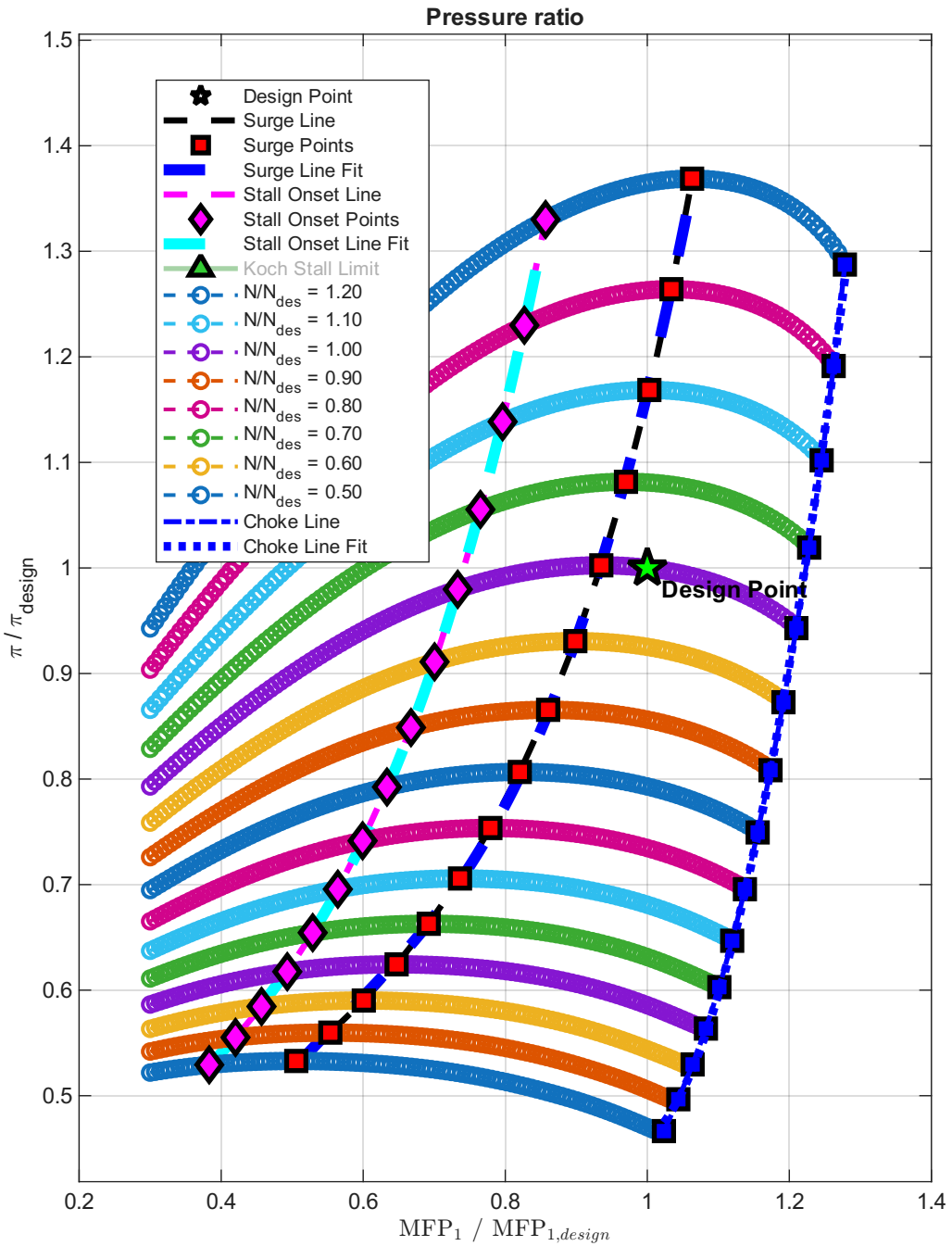


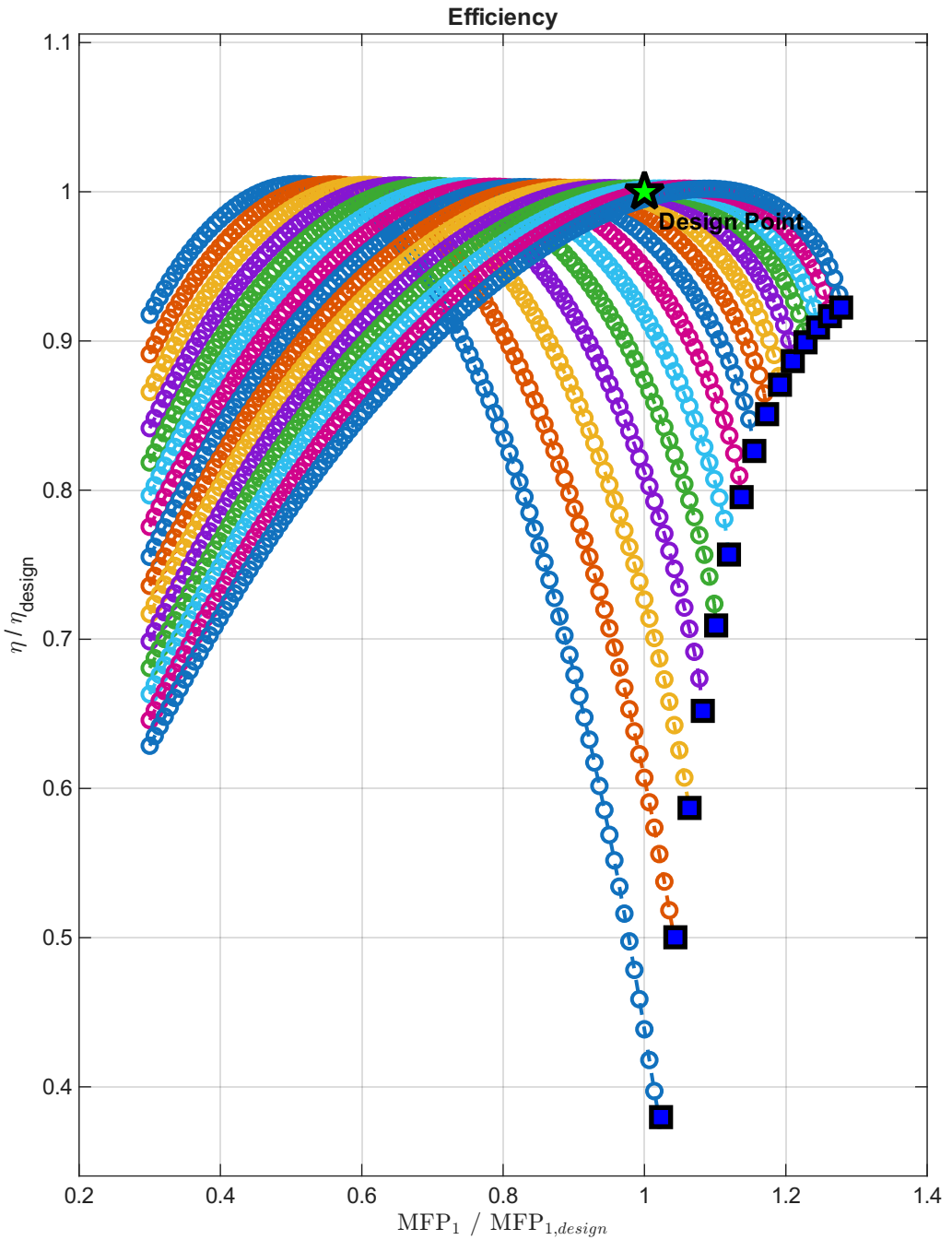
Non-dimensional map - Rotor-Rotor Different Shaft



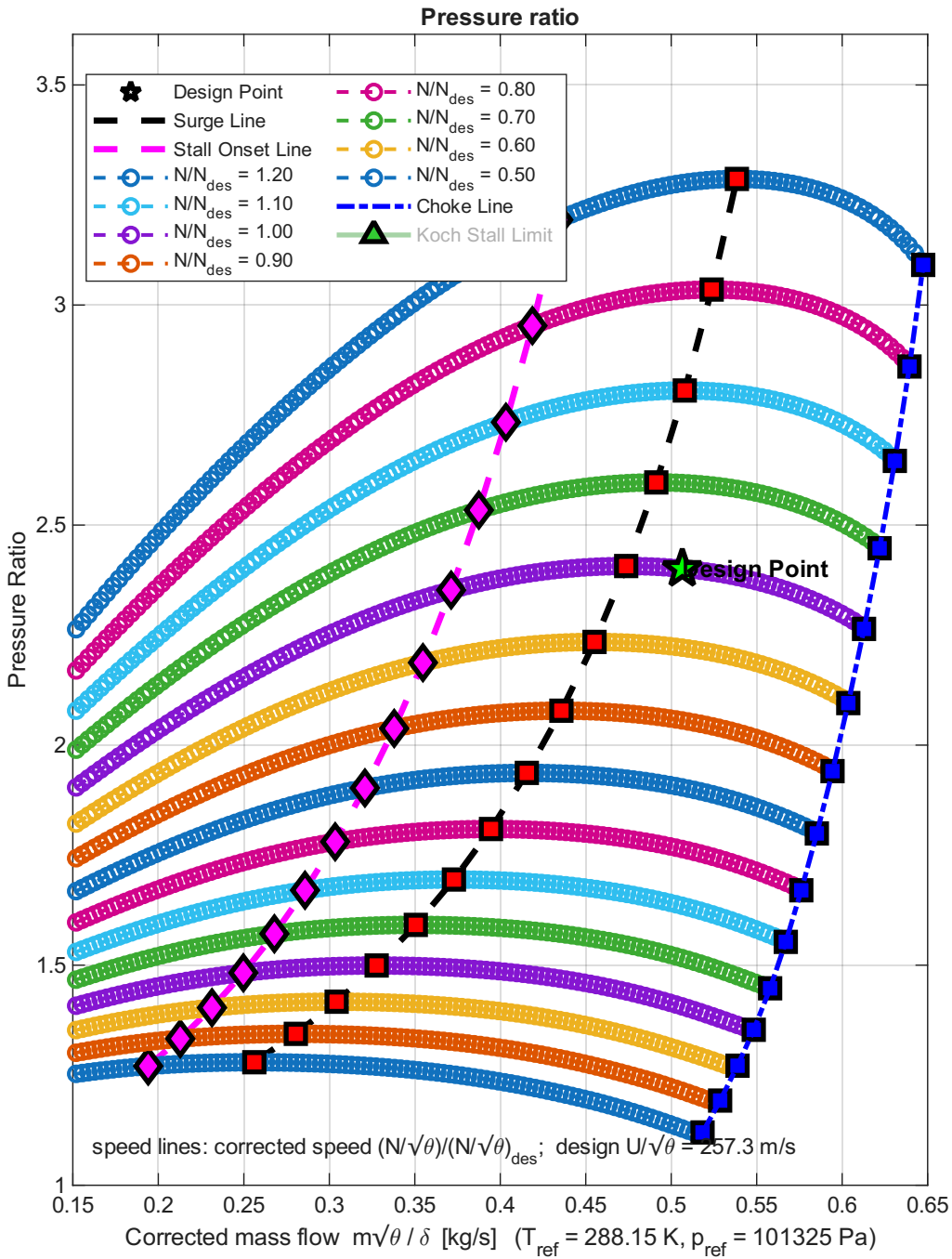


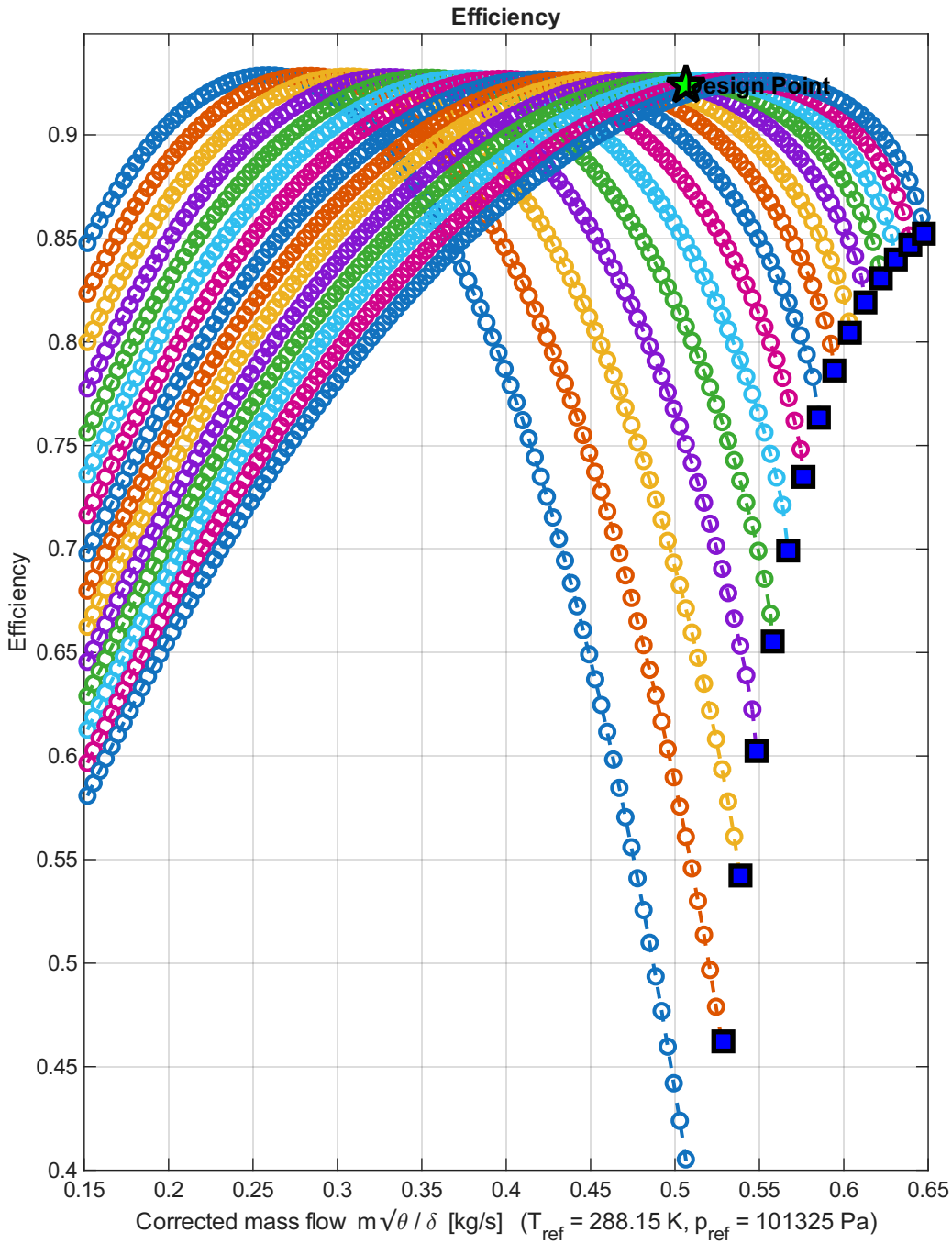
Relative non-dimensional map - Rotor-Rotor Different Shaft





Corrected map - Rotor-Rotor Different Shaft





Fitted boundary equations (surge, stall onset, choke)

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==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Different Shaft ====
Surge line:
Surge_PR = 66.0581*MFP^3 + -59.0626*MFP^2 + 20.7641*MFP + -1.3077   [R^2 = 0.99978]
Stall-onset line (Howell):
StallOnset_PR = 59.5775*MFP^3 + -32.5480*MFP^2 + 9.3428*MFP + 0.2269   [R^2 = 0.99997]
Choke line:
Choke_PR = 77.5786*MFP^2 + -78.2164*MFP + 20.7960   [R^2 = 0.99954]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Different Shaft ====
Surge line:
(PR/PR_design) = 3.8537*x^3 + -6.6346*x^2 + 4.4913*x + -0.5447   (x = MFP/MFP_design)   [R^2 = 0.99978]
Stall-onset line (Howell):
(PR/PR_design) = 3.4756*x^3 + -3.6562*x^2 + 2.0209*x + 0.0945   (x = MFP/MFP_design)   [R^2 = 0.99997]
Choke line:
(PR/PR_design) = 8.7146*x^2 + -16.9184*x + 8.6616   (x = MFP/MFP_design)   [R^2 = 0.99954]
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Design point

Quantity	Value
Configuration	Rotor-Rotor Different Shaft
Pressure ratio PR_des	2.4009
Isentropic efficiency eta_des	0.92395
Mass flow m_des [kg/s]	0.5
MFP_1,des	0.51933
Corrected mass flow [kg/s]	0.50649
Inlet Tt1 [K] / Pt1 [Pa]	288 / 100000
Correlation set	polynomial
Howell factor	2

Choke

N/N_des	m_choke [kg/s]	MFP_1	PR	Eff	Choking row	Inside compressor region (PR>1)
0.5	0.51171	0.5315	1.12	0.351	Rotor 2 exit (relative)	yes
0.55	0.52186	0.542	1.192	0.462	Rotor 2 exit (relative)	yes
0.6	0.5319	0.5525	1.272	0.542	Rotor 2 exit (relative)	yes
0.65	0.54128	0.5622	1.353	0.602	Rotor 1 exit (relative)	yes
0.7	0.55073	0.572	1.447	0.655	Rotor 1 exit (relative)	yes
0.75	0.55978	0.5814	1.553	0.699	Rotor 1 exit (relative)	yes
0.8	0.56873	0.5907	1.67	0.735	Rotor 1 exit (relative)	yes
0.85	0.57771	0.6	1.799	0.763	Rotor 1 exit (relative)	yes
0.9	0.58676	0.6094	1.941	0.786	Rotor 1 exit (relative)	yes
0.95	0.59586	0.6189	2.096	0.804	Rotor 1 exit (relative)	yes
1	0.60496	0.6283	2.264	0.819	Rotor 1 exit (relative)	yes
1.05	0.61396	0.6377	2.447	0.831	Rotor 1 exit (relative)	yes
1.1	0.62275	0.6468	2.646	0.84	Rotor 1 exit (relative)	yes
1.15	0.63121	0.6556	2.86	0.847	Rotor 1 exit (relative)	yes
1.2	0.63913	0.6638	3.091	0.852	Rotor 1 exit (relative)	yes

Stall onset (Howell)

N/N_des	m_stall [kg/s]	PR	Eff	Stalling row
0.5	0.19158	1.271	0.9	Rotor 1
0.55	0.21009	1.333	0.899	Rotor 1
0.6	0.22841	1.403	0.899	Rotor 1
0.65	0.24654	1.483	0.898	Rotor 1
0.7	0.26445	1.571	0.897	Rotor 1
0.75	0.28213	1.67	0.897	Rotor 1
0.8	0.29957	1.78	0.896	Rotor 1
0.85	0.31674	1.903	0.895	Rotor 1
0.9	0.33363	2.038	0.894	Rotor 1
0.95	0.35024	2.187	0.893	Rotor 1
1	0.36654	2.352	0.893	Rotor 1
1.05	0.38252	2.534	0.892	Rotor 1
1.1	0.39816	2.734	0.891	Rotor 1
1.15	0.41346	2.953	0.89	Rotor 1
1.2	0.4284	3.194	0.889	Rotor 1

Peak (zero slope)

N/N_des	m_peak [kg/s]	PR	Eff	m_stall - m_peak [kg/s]	First limit on throttling
0.5	0.25286	1.279	0.929	-0.06128	peak (zero slope)
0.55	0.27683	1.344	0.928	-0.06675	peak (zero slope)
0.6	0.30041	1.417	0.928	-0.07199	peak (zero slope)
0.65	0.32352	1.499	0.928	-0.07698	peak (zero slope)
0.7	0.34612	1.591	0.928	-0.08167	peak (zero slope)
0.75	0.36814	1.695	0.927	-0.08601	peak (zero slope)
0.8	0.38954	1.809	0.927	-0.08997	peak (zero slope)
0.85	0.41024	1.937	0.927	-0.0935	peak (zero slope)
0.9	0.43019	2.078	0.927	-0.09655	peak (zero slope)
0.95	0.44933	2.234	0.926	-0.09909	peak (zero slope)
1	0.46761	2.407	0.926	-0.10107	peak (zero slope)
1.05	0.485	2.597	0.926	-0.10248	peak (zero slope)
1.1	0.50146	2.806	0.926	-0.1033	peak (zero slope)
1.15	0.51697	3.035	0.925	-0.10351	peak (zero slope)
1.2	0.53154	3.287	0.925	-0.10314	peak (zero slope)

Koch limit

N/N_des	m_Koch [kg/s]	PR	Eff	Limiting stage
0.5	0.23215	1.278	0.925	Rotors R1+R2
0.55	0.25739	1.343	0.926	Rotors R1+R2
0.6	0.28334	1.416	0.926	Rotors R1+R2
0.65	0.30995	1.499	0.927	Rotors R1+R2
0.7	0.33743	1.591	0.927	Rotors R1+R2
0.75	0.36554	1.695	0.927	Rotors R1+R2
0.8	0.39409	1.809	0.927	Rotors R1+R2
0.85	0.42293	1.936	0.927	Rotors R1+R2
0.9	0.45169	2.076	0.926	Rotors R1+R2
0.95	0.4803	2.23	0.924	Rotors R1+R2
1	0.5086	2.397	0.922	Rotors R1+R2
1.05	0.53624	2.579	0.919	Rotors R1+R2
1.1	0.56286	2.774	0.915	Rotors R1+R2
1.15	0.58796	2.983	0.91	Rotors R1+R2
1.2	0.61071	3.203	0.903	Rotors R1+R2

Stability limit & Greitzer B

N/N_des	First limit	m_lim [kg/s]	PR	U [m/s]	a_p [m/s]	U/(2a_p)	(V_p/(A_c L_c))_crit	B	Beyond the limit
0.5	peak	0.25286	1.279	128.6	353.3	0.1821	19.31	NaN	-
0.55	peak	0.27683	1.344	141.5	356	0.1988	16.2	NaN	-
0.6	peak	0.30041	1.417	154.4	358.8	0.2151	13.83	NaN	-
0.65	peak	0.32352	1.499	167.2	362	0.231	11.99	NaN	-
0.7	peak	0.34612	1.591	180.1	365.3	0.2465	10.53	NaN	-
0.75	peak	0.36814	1.695	193	368.8	0.2616	9.35	NaN	-
0.8	peak	0.38954	1.809	205.8	372.5	0.2763	8.39	NaN	-
0.85	peak	0.41024	1.937	218.7	376.4	0.2905	7.58	NaN	-
0.9	peak	0.43019	2.078	231.6	380.5	0.3043	6.91	NaN	-
0.95	peak	0.44933	2.234	244.4	384.7	0.3177	6.34	NaN	-
1	peak	0.46761	2.407	257.3	389.1	0.3306	5.85	NaN	-
1.05	peak	0.485	2.597	270.1	393.6	0.3432	5.43	NaN	-
1.1	peak	0.50146	2.806	283	398.2	0.3553	5.07	NaN	-
1.15	peak	0.51697	3.035	295.9	403	0.3671	4.75	NaN	-
1.2	peak	0.53154	3.287	308.7	407.9	0.3785	4.47	NaN	-

Blade rows

Operating point (N = 1)	m [kg/s]	Blade row	Incidence i [deg]	(i - i_des)/deflection	Loss omega	omega/omega_des	M_in (row frame)	M_out (row frame)	de Haller V_out/V_in	Deviation [deg]	Watch
Design point	0.5	Rotor 1	5.5	0	0.1186	1	0.75	0.497	0.682	5	de Haller < 0.72
Design point	0.5	Rotor 2	3.75	0	0.1108	1	0.619	0.458	0.753	8.53	
First limit (peak)	0.46761	Rotor 1	8.7	0.112	0.1262	1.064	0.736	0.449	0.629	5.48	de Haller < 0.72
First limit (peak)	0.46761	Rotor 2	5.63	0.036	0.1116	1.007	0.574	0.414	0.732	8.73	
Near choke (0.99 m_choke)	0.59891	Rotor 1	-5.76	-0.394	0.2116	1.783	0.83	0.848	1.018	5	
Near choke (0.99 m_choke)	0.59891	Rotor 2	-2.78	-0.127	0.1209	1.09	0.964	0.714	0.769	8.32	M_in > 0.9